

**Bureau of Indian Standards**

Manak Bhawan, 9, Bahadur Shah Zafar Marg  
New Delhi - 110002

**IS 2062:2011**

2011

**STEEL FOR GENERAL STRUCTURAL PURPOSES**

**SPECIFICATION**

ICS 77.140.01

BIS 2011

*Indian Standard*

## 1 SCOPE

This standard covers the requirements for structural steel for general structural and engineering purposes in the form of plates, sections (including hollow sections), bars, and wire rods. The standard specifies chemical composition and mechanical properties for various grades.

## 2 GRADE DESIGNATION

Structural steel shall be designated by the minimum yield stress in MPa. The following grades are covered: E165 (Fe 410 W A), E235 (Fe 410 W B), E250 (Fe 410 W C), E275 (Fe 410 W D), E300, E350, E410, E450.

## 3 CHEMICAL COMPOSITION

The steel shall conform to the chemical composition given in Table 1.

| Grade             | C %, max | Mn %     | S %, max | P %, max | Si %     |
|-------------------|----------|----------|----------|----------|----------|
| E250 (Fe 410 W C) | 0.22     | 1.50 max | 0.045    | 0.045    | 0.40 max |
| E300              | 0.22     | 1.50 max | 0.045    | 0.045    | 0.40 max |
| E350              | 0.20     | 1.60 max | 0.040    | 0.040    | 0.40 max |
| E410              | 0.20     | 1.60 max | 0.040    | 0.040    | 0.40 max |

## 4 MECHANICAL PROPERTIES

| Grade             | Yield Stress MPa, min | Tensile Strength MPa | Elongation %, min |
|-------------------|-----------------------|----------------------|-------------------|
| E250 (Fe 410 W C) | 250                   | 410-540              | 23                |
| E300              | 300                   | 440-570              | 22                |
| E350              | 350                   | 490-610              | 20                |
| E410              | 410                   | 540-660              | 18                |

## 5 DIMENSIONAL TOLERANCES

### 5.1 Plate Thickness

For plates of thickness up to 20 mm, the tolerance shall be 0.5 mm. For plates of thickness above 20 mm up to 40 mm, the tolerance shall be 0.75 mm.

## 6 TEST METHODS

- Tensile Test: IS 1608 (Part 1) - Room Temperature Test
- Chemical Analysis: IS 228 (Part 1 to 12)
- Bend Test: IS 1599
- Dimensional Check: IS 1852

## 7 MARKING

Each plate, section, or bar shall be legibly marked with: (a) Manufacturer's name or trademark, (b) Standard designation IS 2062, (c) Grade of steel, (d) Heat number, (e) Size dimensions.